Experiment 21: Perform Sharpening of Image using Laplacian Mask implemented with an extension of diagonal neighbors

AIM

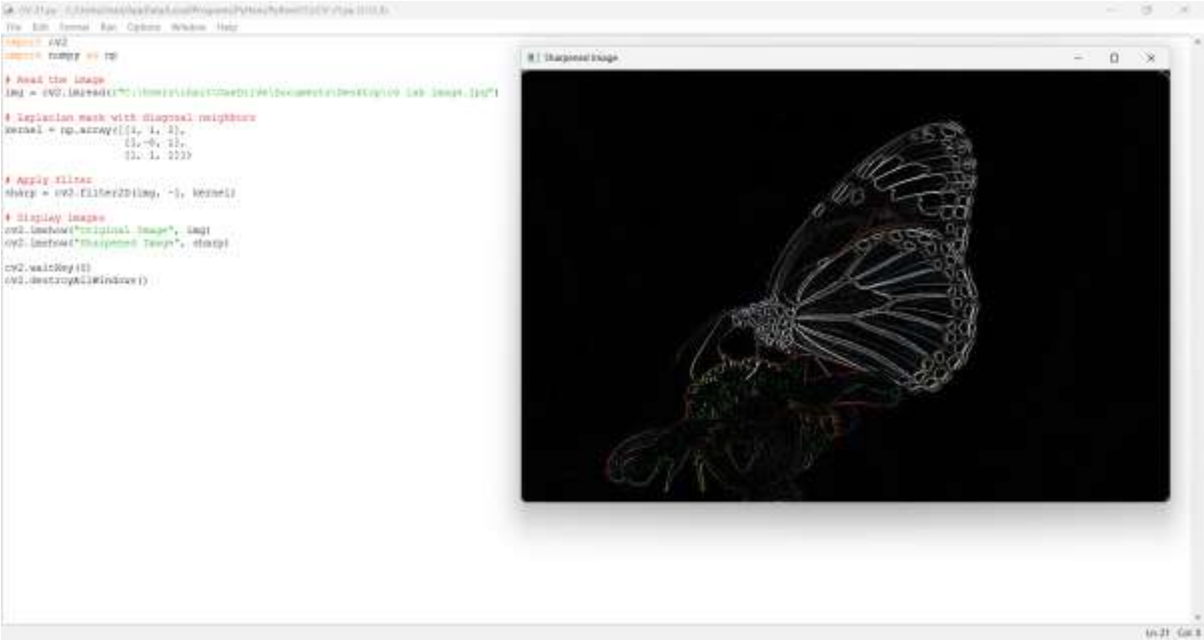
To sharpen an image using a Laplacian mask with diagonal neighbors.

PROGRAM

```
import cv2
import numpy as np

# Read the image
img = cv2.imread(r"C:\Users\chait\OneDrive\Documents\Desktop\cv lab image.jpg")

# Laplacian mask with diagonal neighbors
kernel = np.array([[1, 1, 1],
                   [1,-8, 1],
                   [1, 1, 1]])

# Apply filter
sharp = cv2.filter2D(img, -1, kernel)

# Display images
cv2.imshow("Original Image", img)
cv2.imshow("Sharpened Image", sharp)

cv2.waitKey(0)
cv2.destroyAllWindows()
```

OUTPUT

The image is sharpened using the Laplacian mask with diagonal neighbors.



Explanation

We applied a Laplacian mask that considers both horizontal, vertical, and diagonal neighboring pixels. This enhances image edges and fine details, producing a sharper image.